What is CSS3?

- Cascading Style Sheets Level 3.
- It is a style sheet language used to describe the presentation, formatting and layout of a document written in markup language (HTML).
- It is a language designed to simplify the process of making web pages presentable.
- HTML tags create the content and structure but without CSS, the content appears in default style that lacks appeal.

CSS	CSS3
Capable of positioning text and objects	Capable of making web pages more attractive and takes less time to create
Cannot be split into modules	Can be broken down into modules.
Slower compared to CSS3	Faster than CSS
Cannot build 3Da animation and transformation	Supports animation and 3D transformation

Advantages of CSS

Consistent Design; By using CSS, we can use same design across an entire website or on various web pages. This reduces the need of rewriting the same code multiple times. User can create an external CSS and link it to various web pages for uniform and consistent design.

Easy Maintenance; By using CSS, it is easy to maintain websites, as with CSS, change in single line of code affects the entire webpage so that we don't have to change each element individually.

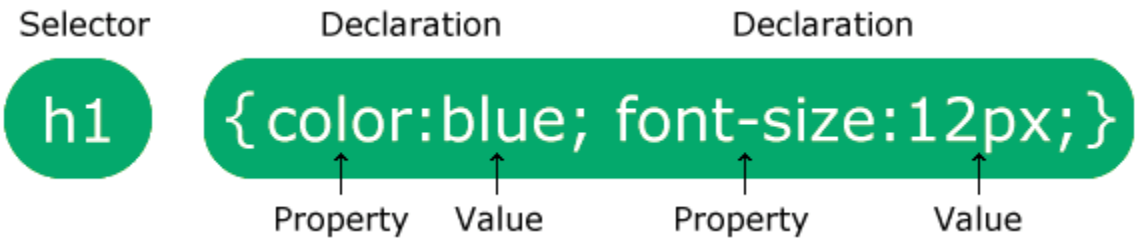
Content Separation; CSS can be used 3 ways i.e. Inline, Internal and External. External CSS allows user to separate HTML and CSS which increases the readability of code.

Faster Loading Time; CSS improves load time as it's cached by browser once CSS is downloaded. It's not needed to download CSS again on subsequent visits making webpage to load faster.

Animations and Transitions; CSS allows animations and transitions in a web page improving user experience, making web page more interactive without using any scripting language.

CSS Syntax

- A CSS rule consists of a selector and a declaration block.



- The selector points to the HTML element you want to style.
- The declaration block contains one or more declarations separated by semicolon.
- Each declaration includes a CSS property name and a value, separated by colon.
- Multiple CSS declarations are separated by semicolon and declaration blocks are surrounded by curly braces.
- Example

```
p
{
        color: red;
        text-align: center;
}
```

- In above example, all <p> elements will be center-aligned with a red text color.
- p is a selector.
- Color is a property and red is the property value. Text-align is a property and center is a property value.

Selectors in CSS

- Selectors define which HTML elements are styled.
- CSS offers several types of selectors.
- Following are the types of selectors.

  Universal Selector
- Applies styles to all elements.
- Following is an example.

```
*{
```

```
        margin: 0;
        padding: 0;
}
```

Type Selector

- Targets specific HTML elements.
- Following is an example.

```
h1
{
        color: red;
        text-align: right;
}
```

Class Selector

- Styles elements with a specific class attribute.
- Following is an example.

```
.box
{
        border: 2px;
        padding: 10px;
}
```

ID Selector

- Targets a single element with a specific ID.
- Following is an example.

```
#header
{
        background-color: lightgray;
}
```

External Style Sheet

- The most common and efficient method, where styles are defined in a separate .CSS file and linked to multiple HTML pages. Changing a single CSS file updates the entire website's appearance.
- To use external CSS, you need two files: HTML and CSS.

  1. Create the CSS file

     Create a new file using any text editor and save it with .CSS extension. This file should contain only CSS rules and no HTML tags. Following is an example.

```
body
{
        background-color: yellow;
}
h1
{
        color: maroon;
        margin-left : 40px;



}
```

2. <u>Link CSS file to HTML document</u>

In your HTML file, use the <link> element inside <head> tag to reference the external CSS file. Following is an example.

```
<!DOCTYPE HTML>
<HTML>
<HEAD>
        <TITLE> External CSS Example </TITLE>
        <LINK rel="stylesheet" href="style.CSS">
</HEAD>
<BODY>
        <H1> This is a heading </H1>
</BODY>
</HTML>
```

The <link> tag uses two required attributes.

rel="stylesheet"; defines the relationship between the HTML document and the linked file, indicating it is stylesheet.

href="style.CSS"; specifies the path of an external CSS file. If the CSS file in a different folder, you would use an entire path of the file.

<u>Internal Style Sheet</u>

- The style rules are placed within a <style> element in the <head> section of HTML file.
- This is useful for page specific style.
- Following is an example.

```
<!DOCTYPE HTML>
```

```
<HEAD>
<TITLE> Internal CSS example </TITLE>
<style>
        body
        {
                background-color: yellow;
        }
        h1
        {
                color: maroon;
                margin-left : 40px;
        }
</style>
</HEAD>
<BODY>
        <H1> This is a heading </H1>
</BODY>
<HTML>
```

Inline Style Sheet

- Styles are applied directly to specific HTML element using the <style> attribute.
- This is useful for unique changes to a single element.
- Following is an example.

```
<!DOCTYPE HTML>
<HEAD>
        <TITLE> Example of Inline CSS </TITLE>
</HEAD>
<BODY>
        <H1 style="color:maroon ; text-align: center;"> This is a
        heading</H1>
        <p style="color:red ; font-size:18px;"> This is a paragraph </p>

</BODY>
</HTML>
```

CSS Color

- In CSS, colors are specified by using a predefined color name, RGB (Red Green Blue, values (0 to 255)), RGBA (Red Green Blue Alpha to control opacity, with a value between 0.0 (fully transparent) and 1.0 (fully opaque)), HEX (Hexadecimal value preceded by (#) ranging from 00 to FF), HSL (Hue, Saturation and Lightness) and HSLA (includes alpha channel for opacity).
- CSS/HTML support 140 color names.
- CSS colors are used to change text color (color: red), background color (backgrounnd-color: orange) or border color (border-color: yellow).

CSS Background

- CSS background properties are used to add background effects for elements.
- Following are the properties of CSS background.
- background-color; Defines the background color of an element using color names, RGB, HEX or HSL values.
- background-image; Adds one or more background images of an element.
- background-repeat; Specifies the how the background image repeats. Common values include repeat (default, repeats horizontally and vertically), repeat-x (horizontally only), repeat-y (vertically only), no-repeat (once).
- background-position; Sets the starting position of the background image within the element. Values can be left top, left center, left bottom, center top, center center, center bottom, right top, right center or right bottom.
- background-size; Specifies the size of the background image. Values can be auto (default); the background image is displayed in its original size, contain; will resize the background image to make sure it remains fully visible, cover; will resize the background image to make sure the element is fully covered. Examples. background-size: cover, 150px, 100px;, background-size: 150px ;(Sets the width to 150px, height is auto), background-size:50% ; (Sets width to 50% of container, height is auto).
- background-attachment; Determines whether the background image scrolls with the page or stays in a fixed position. Values are scroll (default), fixed or local (scroll with the element's content).
- background-origin; Specifies the positioning area of the background image(s) within an element's box model. This property has no effect if background-attachment is "fixed". This property can have one of the following values.

padding-box; The background image starts from the top-left corner of the padding edge. This is default.
border-box; The background image starts from the top-left corner of the border edge.
content-box; The background image starts from the top-left corner of the content edge.

CSS Border

- A border is a rectangular line drawn along the 4 sides of an HTML element.
- Following are the properties of border.
- border-style; specifies the type of border to display. Possible values are dotted (defines a dotted border), dashed (defines a dashed border), solid (defines a solid border), double (defines a double border), groove (defines a 3D grooved border. The effect depends on the border color value), ridge (defines a 3D ridged border. The effect depends on border-color value), inset (defines a 3D inset border. The effect depends on the border-color value), outset (defines a 3D outset border. The effect depends on border-color value), none (defines no border).
- border-width; specifies the thickness of the border. Possible values are thin, medium (default), thick or a specific length unit in px. For example:

```
h2
 {
        border-width: 2px;
}
```

- border-color; defines the color of the border. Value can be color name (e.g. red), HEX value, RGB value or HSL value. The border-color is a shorthand for border-top-color, border-right-color, border-bottom-color and border-left – color. Specifying border-color: red; (All four sides are red), border-color: red green; (Top and bottom borders are red, left and right borders are green), border-color: red green blue; (Top is red, left and right are green, bottom is blue), border-color: red green blue yellow; (Top is red, right is green, bottom is blue and left is yellow).
- border; A shorthand property for setting width, style and color in a single declaration.

- Indivijdual side shorthand; Shorthand also exist for each side, combining the width, style and color for the specific edge. (e.g. border-top, border-right, border-bottom, border-left). For example:

```
h2
{
        border-bottom: 2px dashed blue;
}
```

- border-radius; adds rounded corner to an element's border. For example:

```
h1
{
        border-radius: 10px;
}
```

- border-image; Allows using an image as the border around element. An element must have a standard border property set for border-image to work. The most efficient way to use the border-image shorthand property: border-image: source slice width outset repeat.

  border-image-source property specifies the path of an image to be used as a border (instead of normal border around an element). Following is an example.

```
h1
{
        border-image-source: url ('border.png');
}
```

  border-image-slice property specifies how to slice the image specified by border-image-source. The image is always sliced into nine sections: four corners, four edges and middle. Following is an example.

```
h1
{
        border-image-slice: 30%;
}
```

  border-image-width property specifies the width of the border image. Following is an example.

```
h1
{
        border-image-source: url('border.png');
        border-image-width: 10px;
```

```
}
```

border-image-outset property specifies the amount by which border image area extends beyond the border box. An image source must be defined using border-image-source (or the border-image shorthand) for the outset to have a visible effect. Following is an example.

```
h1
{
        border-image-source: url ('border.png');
        border-image-outset: 10px;
}
```

border-image-repeat property specifies whether the border image should be repeated, rounded, spaced or stretched.

Stretch; This is the default value. It stretches the source image's edge regions to fill the entire gap between the corners of the border. This works best with simple, flat or smooth images that can be scaled without degrading their appearance.

Repeat; This value repeats the image to fill border area. If the image does not fit a whole number of times, the final tile may be clipped to achieve a proper fit. This is suitable for creating a traditional, tiled pattern effect.

Round; This is similar to repeat but it rescales the image slightly so that a whole number of tiles fit perfectly into the available space, ensuring no clipping or gaps occur. This is ideal for continuous, unbroken patterns.

Space; This value also tiles the image without clipping but instead of rescaling the image, it distributes any extra space evenly between the tiles to fill the border area.

- border-collapse; A property used specifically with HTML tables to determine if border should be collapsed into a single line or remains separate.

CSS Fonts

- CSS fonts are used to style and enhance the appearance of text on web page.
- It is used to set the size, weight, style and color of text.
- Following are the properties of CSS fonts.
- font-family; It defines which font(s) should be used for text. You can provide a list of fonts as a fallback mechanism in case the preferred font is unavailable.

  Syntax: font-family: "Font Name", generic-font-name;

Example:

```
body
{
        font-family: “Arial”, sans-serif;
}
```

In the above example, the browser will first try to display the text in Arial. If Arial is not available, it will fall back to sans-serif font.

- font-size; It controls the size of the text. You can set size in pixels. Syntax: font-size: value;

  Example:

  ```
  h1
  {
          font-size: 26px;
  }
  ```

- font-weight; Specifies the thickness (boldness) of characters. Values can be keywords (normal, bold, lighter, and bolder) or numeric values from 100 to 900. 100 is for light and 900 for extra bold. Syntax: font-weight: value;

  Example:

  ```
  p
  {
          font-weight: bold;
  }
  strong
  {
          font-weight: 700;
  }
  ```

- font-style; Defines the style of the font. Values can be italic or normal. Syntax: font-style: value;

  Example:

  ```
  p
  {
          font-style: italic;
  }
  ```

- font-variant; It specifies if the fonts to be displayed in normal or small caps. Small caps are displayed when all the letters of the word are in capitals with

uppercase characters slightly larger than other letters. Syntax: font-variant: value;

Example:

```
h1
{
        Font-variant: small-caps;
}
```

- line-height; It defines the space between lines of text. Syntax: line-height: value;

  Example:

```
p
{
        line-height: 1.5;
}
```

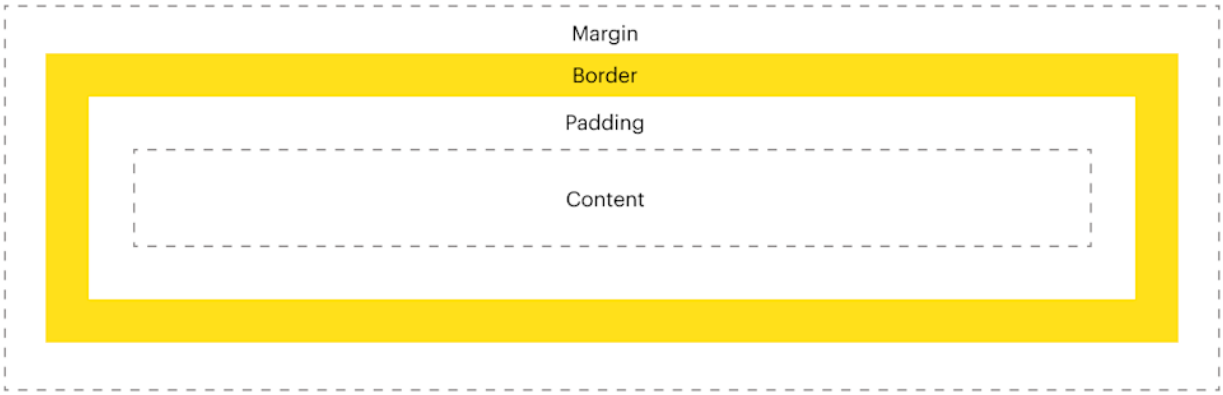
CSS Margin

- CSS margin properties are used to create space around elements, outside of any defined borders.
- Margins define the distance between an element's border and surrounding elements.
- CSS has following properties for specifying the margin for each side of an element.
- margin-top; sets the top margin of an element.
- margin-right; sets the right margin of an element.
- margin-bottom; sets the bottom margin of an element.
- margin-left; sets the left margin of an element.
- margin; A shorthand property to define the margin of an element.
- Following are the examples.
- If the margin property has four values like margin: 10px 15px 20px 25px; It specifies top margin is 10px, right margin is 15px, bottom margin is 20px and left margin is 25px.
- If the margin property has three values like: margin: 10px 15px 20px; It specifies top margin is 10px, right and left margins are 15px and bottom margin is 20px.
- If the margin property has two values like margin: 10px, 15px; It specifies top and bottom margins are 10px and right and left margins are 15px.

- If the margin property has one value like margin: 10px; It specifies all four margins are 10px.

CSS Padding

- An element's padding area is the space between its content and its border.
- CSS padding is the space inside an element between its content and border.



- Padding creates extra space within an element and margin creates extra space outside an element.
- You can control padding on each side of an element using following properties.
- padding-top; sets top padding.
- padding-right; sets the right padding.
- padding-bottom; sets the bottom padding.
- padding-left; sets the padding.
- padding; A shorthand property to define the padding of an element.
- Following are the examples.
- If the padding property has four values like padding: 10px 15px 20px 25px; It specifies top padding is 10px, right padding is 15px, bottom padding is 20px and left padding is 25px.
- If the padding property has three values like: padding: 10px 15px 20px; It specifies top padding is 10px, right and left paddings are 15px and bottom padding is 20px.
- If the padding property has two values like padding: 10px, 15px; It specifies top and bottom paddings are 10px and right and left paddings are 15px.
- If the padding property has one value like padding: 10px; It specifies all four paddings are 10px.

CSS Images

- CSS images help enhance appearance on a web page.
- It allows us to adjust size, shape, border and effects making images look more attractive.
- Following are the properties of CSS image.
- Width and height; used to set width and height of an image. If you set a width and height that is different from the image's own width and height, then browser will scale the image up or down depending on the value you specify. Example:

```
img
{
        width: 300px;
        height: 250px;
}
```

- border-radius; Adds rounded corners to an image.

  Example:

```
Img
{
        Border-radius: 4px;
}
```

- opacity; Specifies the transparency of the entire image with values ranging from 0.0 (fully transparent) to 1.0 (fully opaque).
- filter; Applies visual effects like blur (), grayscale (), sepia(), brightness() or contrast(). Blur function applies a blur effect to an image. A larger value will create more blur. Example:

```
img
{
        filter: blur(6px);
}
```

  Grayscale() filter function converts an image to grayscale. 100% will make the image completely grayscale. 0% will have no effect. Example:

```
img
{
        filter:grayscale(60%);
}
```

Sepia() filter function converts an image to a sepia tone (more brown/yellow) color. 0% will have no effect. 100% applies full sepia effect. Example:

```
img
{
	Filter: sepia (60%);
}
```

brightness() filter function adjusts the brightness of an image. 100% is default and represents the original brightness. Values over 100% will provide brighter result. Values under 100% will provide darker result. 0% will make the image completely black. Example:

```
img
{
	Filter: brightness(150%);
}
```

Contrast() filter function adjusts the contrast of an image. 100% is default and represents original contrast. Values over 100% increases the contrast. Values under 100% decreases the contrast. 0% makes an image completely gray. Example:

```
img
{
	filter: contrast(50%);
}
```

- <u>max-width and max-height</u>; are commonly used on images to control how large they are allowed to get, without forcing a fixed size.